Predictive Enrichment of Geometry-Induced Soft-State Abstractions: Information Retention and Decoder Access

Anonymous authors
Paper under double-blind review

Abstract

Geometry-induced soft-state abstractions describe an observed history through relative affinities to reference regions. Their geometric interpretation does not establish that they retain distinctions needed for forecasting or make those distinctions accessible to a restricted predictor. We investigate this gap using Kernel Affine Hull Machines (KAHMs) as the principal reconstruction method. We distinguish information discarded by score normalization, local-position information, and access through a finite decoder. Explicit inverse relations show how partition normalizers recover the reconstruction-score vector and give conditions under which gated local coordinates preserve the input. A separate comparison of matched raw reconstruction spaces clarifies why information retention, linear expressivity, and dynamical closure are different properties. We evaluate KAHM, affine, and positive-kernel reconstructions with linear ridge and finite nonlinear decoders, using validation-selected native dimensions and 64 independent repetitions of each of four simulation laws. In a prespecified family of 40 paired comparisons, local-position enrichment lowers KAHM membership-only loss in six of eight law–decoder comparisons. Advantages over alternative constructions are conditional: none of the four nonlinear-decoder comparisons against matched positive-kernel reconstruction resolves the ordering, and several comparisons favor affine reconstructions or raw histories. The findings explain why geometric fidelity alone is insufficient for selecting predictive features and why enrichment must be assessed with a specified decoder and common target. They support conditional predictive usefulness, not universal KAHM superiority or autonomous soft-state closure.

1 Introduction

Forecasting requires a representation that preserves distinctions relevant to future observations and allows a predictive model to use them. Geometry-induced soft-state abstractions describe an input through graded relationships with reference regions. When these relationships are obtained from reconstruction scores, they have a direct geometric interpretation: a reference set receives greater affinity when it reconstructs the query more successfully. That interpretation alone does not explain whether the resulting coordinates support prediction of an observed future response.

This gap motivates the present study. Reducing a reconstruction vector to a score can discard position within a reference region, and normalizing scores into memberships removes their overall affinity scale. Conversely, an enriched encoding can preserve the input while requiring an inverse that a chosen linear or finite nonlinear decoder does not implement well. Information loss in the encoding and limited access through the fitted decoder are therefore distinct obstacles. Distinguishing them matters when deciding whether to enrich a representation, change the readout, or prefer a simpler construction.

We investigate these obstacles using Kernel Affine Hull Machine (KAHM) reconstruction scores as a concrete family of soft coordinates (Kumar et al., 2024; 2025; Kumar & Kargaran, 2026). The central question is which distinctions the coordinates retain, what enrichment restores, and under which specified representation–decoder procedures prediction improves. We compare memberships alone, memberships with partition log

normalizers, memberships with gated local principal coordinates, and the full score vector. Normalizers retain absolute affinity information, whereas local coordinates retain position; these are parallel enrichments rather than successive stages of one hierarchy. Matched affine and positive-kernel reconstructions test whether an observed benefit is specific to KAHM, and raw histories provide a separate forecasting control. All procedures predict the same observed response.

The analysis characterizes score recovery, input recovery, and conditioning. It also gives a full-rank affine example in which reconstruction is exact but memberships carry no position information. The experiment tests complete representation–decoder procedures with matched reconstruction controls and validation-selected dimension, retaining all four laws and both decoders in one primary family. A separate raw-span comparison describes a limited implication for candidate Koopman observable spaces; it is not a closure theorem for the nonlinear score encodings.

The empirical findings separate enrichment from reconstruction-specific advantage. Local position improves KAHM memberships in six of eight law–decoder comparisons, but all four nonlinear-decoder comparisons with positive-kernel reconstruction remain unresolved. Unfavorable linear-autoregression and Duffing results are retained.

Relation to prior constructions. Local prediction from delay coordinates predates the present representation (Farmer & Sidorowich, 1987). Local coordinate coding and local tangent coding motivate localized weighted displacements (Yu et al., 2009; Yu & Zhang, 2010); the local-position block is an application of that established design. The distinction between information and access through a restricted predictor is also central to usable-information theory (Xu et al., 2020). We do not claim a new general information principle or an optimal-decoder estimate. The positive-kernel reconstruction is a normalized vector-valued kernel smoother (Nadaraya, 1964), while normalized inverse-kernel constructions have an antecedent in rescaled interpolation (De Marchi et al., 2017). The nonlinear decoder is a finite Nyström approximation with an additional linear input, rather than unrestricted kernel regression (Williams & Seeger, 2001).

Distinction from fixed-abstraction certification and dynamical learning. Kumar & Kargaran (2026) study lower prediction-error certificates for a fixed soft map and a prescribed norm-bounded linear predictor class, together with the conditional interpretation of exact coordinate propagation. Our target is instead a common future observation, and our comparison varies the representation, dimension selection, and fitted decoder. Those successor-coordinate certificates do not imply improved external-response prediction. Methods such as VAMPnets and learned Koopman embeddings explicitly model latent evolution (Mardt et al., 2018; Lusch et al., 2018); our direct horizon-specific readouts do not. The present study concerns observable design and predictive access, not an empirical demonstration of autonomous state dynamics.

2 Reconstruction-score representations

2.1 Inputs, notation, and reference geometry

Let (y_t) be a scalar observed series and let $x_t = (y_{t-d+1}, \dots, y_t)^\top \in \mathbb{R}^d$ be the available history. The target is y_{t+h} for a positive horizon h in a finite set $\mathcal{H}$. A single representation supplies separately fitted coefficients for these direct forecasts. A, B, C, and D denote disjoint episode sets used for training, validation, diagnostic evaluation, and terminal evaluation, respectively. No claim that a finite observed history is a Markov state is assumed.

Vectors are columns; $\top$ denotes transpose. The symbols $\|\cdot\|_1$ and $\|\cdot\|_2$ denote the sum of absolute coordinates and the Euclidean norm; $\|\cdot\|_F$ is the Frobenius matrix norm. Logarithms are natural. Also, I_n is the n-dimensional identity matrix, and $\mathbf{1}_n$ is the all-ones vector. The operator col stacks its arguments. Let $z \in \mathbb{R}^d$ be an A-standardized history, using population-divisor standard deviations and replacing a scale at most 10^{-12} by one.

There are B_0 reference partitions. Partition b contains local model indices $\mathcal{J}_b$, with disjoint index sets across partitions, and the total number of local models is $J = \sum_b |\mathcal{J}_b|$. Model j contains n_j reference histories $q_{ji} \in \mathbb{R}^d$ as the rows of $Q_j \in \mathbb{R}^{n_j \times d}$. Its mean is μ_j, and $P_j \in \mathbb{R}^{d \times r_j}$ contains orthonormal principal directions.

References and partitions use A only and are fixed before changing the dimension cap. For cap c, the requested rank starts at $\min\{c, d, n_j - 1\}$. Native positive-variance and projected-range rules can reduce this request; r_j always denotes the actual stored rank. There is no extra $d - 1$ restriction. The supplement specifies the numerical rank policy and its rank-zero case.

2.2 Three matched reconstructions

For positive rank, let S_j be the sample covariance of $P_j^\top q_{ji}$, with divisor $n_j - 1$. The retained positive-variance directions make S_j positive definite. Define

$$\kappa_j(z, z') = \exp\left(-\frac{[P_j^\top(z - z')]^\top S_j^{-1}[P_j^\top(z - z')]}{2r_j}\right). \tag{1}$$

Let $k_j(z) \in \mathbb{R}^{n_j}$ contain the similarities to the references and let G_j be their Gram matrix. A positive regularizer λ_j is fitted to reference reconstruction on A, without response targets. Write $C_j = (G_j + \lambda_j I_{n_j})^{-1}$. The reconstruction rules are

$$\mathcal{A}_j^K(z) = \frac{Q_j^\top C_j k_j(z)}{\mathbf{1}_{n_j}^\top C_j k_j(z)}, \qquad \mathcal{A}_j^R(z) = \frac{Q_j^\top k_j(z)}{\mathbf{1}_{n_j}^\top k_j(z)}, \tag{2}$$
$$\mathcal{A}_j^A(z) = \mu_j + P_j P_j^\top (z - \mu_j).$$

Here K denotes KAHM, R denotes positive-kernel reconstruction, and A denotes affine projection. Their reference histories and bases are matched at a fixed cap; K and R also share the projected Gaussian metric. KAHM weights sum to one when their denominator is nonzero, but can be negative. They are not the soft memberships defined below.

The implementation subtracts the maximum log-kernel entry before exponentiation. This common rescaling leaves the exact normalized quotient unchanged and prevents complete row underflow. A signed-denominator guard requires $|\mathbf{1}^\top C_j k_j| / \|C_j k_j\|_1 \geq 10^{-10}$ when the latter norm is nonzero. A declared invalid query uses the affine reconstruction and recomputes its score; it does not set the score to zero. Rank-zero models use μ_j for all rules. These safeguards define the evaluated numerical map. The exact raw-span analysis concerns the quotient where it is defined, without fallback.

2.3 Scores and soft memberships

For reconstruction rule $m \in \{K, R, A\}$, let $a = \mathcal{A}_j^m(z)$. Define the radial discrepancy $e_j^m(z) = 1 - \exp(-\|z - a\|_2)$ and angular discrepancy $\theta_j^m(z) = \pi^{-1} \arccos(\mathrm{clip}(c(z, a), -1, 1))$, where

$$c(z, a) = \begin{cases} z^\top a / (\|z\|_2 \|a\|_2), & \|z\|_2 \|a\|_2 > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathrm{clip}(u, l, v) = \min\{v, \max\{l, u\}\}.$$

The reconstruction score is

$$T_j^m(z) = \sqrt{\left(e_j^m(z)^2 + \theta_j^m(z)^2\right)/2} \in [0, 1]. \tag{3}$$

This radial-and-angular score follows the KAHM soft-coordinate construction (Kumar & Kargaran, 2026). It is not a forecast error. In particular, its angle is defined relative to the standardized origin. Exact nonzero reconstruction has score zero, while exact reconstruction at $z = 0$ has score $1/(2\sqrt{2})$ under the stated zero-norm convention.

For fixed $\tau > 0$ and $\omega > 0$, normalize separately in each partition:

$$Z_b^m(z) = \sum_{j \in \mathcal{J}_b} (1 - T_j^m(z) + \tau)^\omega, \qquad \phi_j^m(z) = \frac{(1 - T_j^m(z) + \tau)^\omega}{Z_b^m(z)}, \qquad \zeta_b^m(z) = \log Z_b^m(z). \tag{4}$$

The concatenation ϕ^m lies in $\prod_b \Delta_{|\mathcal{J}_b|}$, where $\Delta_q = \{v \in \mathbb{R}^q : v_i \geq 0,\ \mathbf{1}_q^\top v = 1\}$. These memberships describe relative reconstruction affinities, not posterior regime probabilities. Each rule supplies its own scores and memberships. Forcing the three rules to share memberships would remove part of the effect under investigation.

2.4 Four encodings and a raw-input control

Let $T^m = \operatorname{col}_j T_j^m$ and $\zeta^m = \operatorname{col}_b \zeta_b^m$. At a fixed reconstruction and bank, define

$$\begin{aligned} \Gamma_{\mathrm{mem}}^m(z) &= \phi^m(z), & \Gamma_{\mathrm{norm}}^m(z) &= \operatorname{col}(\phi^m(z), \zeta^m(z)), \\ \Gamma_{\mathrm{local}}^m(z) &= \operatorname{col}\big(\phi^m(z), \{\phi_j^m(z) P_j^\top (z-\mu_j)\}_{j=1}^J\big), & \Gamma_{\mathrm{score}}^m(z) &= T^m(z). \end{aligned} \tag{5}$$

Their widths are J, $J+B_0$, $J+\sum_j r_j$, and J, respectively. The standardized history $\Gamma_{\mathrm{raw}}(z) = z$ is a separate control. Enrichment does not retain a hidden raw-history forecast, fit residual targets, or apply a calibrated correction. Only Γ_{mem} has the product-of-simplices interpretation. The normalizer and local encodings need not be nonnegative or normalized.

3 Information retention and candidate observable spaces

The statements in this section condition on the fitted reference geometry. Information recovery concerns exact encodings; it is distinct from a guarantee for a finite fitted decoder. Complete proofs of the propositions are in Appendix A.

3.1 What normalization retains

Proposition 1 (Recovery of the score vector)**.** *Fix $\tau > 0$, $\omega > 0$, and a partition. The pair (ϕ^m, ζ^m) determines its score vector through*

$$T_j^m(z) = 1 + \tau - \exp\left(\frac{\log \phi_j^m(z) + \zeta_{b(j)}^m(z)}{\omega}\right), \tag{6}$$

where $b(j)$ is the partition containing j. Conversely, the scores determine both memberships and normalizers. For a partition with q models,

$$\phi_j^m(z) \geq \frac{\tau^\omega}{\tau^\omega + (q-1)(1+\tau)^\omega} > 0.$$

Thus normalizers recover the full scores, not information already lost when reconstruction vectors were reduced to scores. Memberships alone retain only relative affinities. For example, two distinct constant score vectors give the same uniform memberships but different normalizers. This is an elementary property of the normalization, not a new inverse-learning algorithm.

The exact lower bound is not a practical conditioning guarantee. With $\tau = 10^{-6}$, $\omega = 16$, and four models, it is about 3.33×10^{-97}. Stable logarithmic evaluation preserves the normalization without a small-sum override; it cannot make division by a very small membership well conditioned. Also, an information-equivalent nonlinear reparameterization generally changes linear predictors, Gaussian distances, and isotropic penalties. Equal information does not imply equal fitted forecasts.

3.2 What local coordinates retain

Let $v_j(z) = \phi_j^m(z) P_j^\top (z - \mu_j)$ and let $\mathcal{P} = \operatorname{col}_j P_j^\top \in \mathbb{R}^{(\sum_j r_j) \times d}$.

Proposition 2 (Input recovery and conditioning)**.** *If every membership is positive and $\mathcal{P}$ has column rank d, then $\Gamma_{\mathrm{local}}^m$ is injective. Define*

$$q(z) = \operatorname{col}_j \left(\frac{v_j(z)}{\phi_j^m(z)} + P_j^\top \mu_j\right).$$

The input is then recovered by $z = (\mathcal{P}^\top \mathcal{P})^{-1} \mathcal{P}^\top q(z)$. Moreover, suppose two inputs z, z' satisfy $\phi_j^m(z), \phi_j^m(z') \geq \varepsilon > 0$ and $\|P_j^\top (z' - \mu_j)\|_2 \leq M$ for all j. Writing $v = \operatorname{col}_j v_j$ and letting $\sigma_{\min}(\mathcal{P}) > 0$ be the smallest singular value,

$$\|z - z'\|_2 \leq \frac{\|v(z) - v(z')\|_2 + M \|\phi^m(z) - \phi^m(z')\|_2}{\varepsilon \sigma_{\min}(\mathcal{P})}. \tag{7}$$

The condition is sufficient, not necessary: a rank-deficient stack does not establish information loss because the gates can themselves distinguish inputs. Conversely, full rank does not guarantee a stable inverse, dimensionality reduction, or low fitted error. The local construction can be wider than its input, and the finite decoder need not implement the inverse above.

Exact reconstruction can yield constant memberships. If all affine bases in one partition have rank d, then $\mathcal{A}_j^A(z) = z$ for every model. At a nonzero query all scores are zero; at zero all share the zero-norm score convention. Equation (4) therefore gives uniform memberships at every input. Exact reconstruction by these affine models supplies no distinction through their memberships alone. Their gated full local coordinates, however, retain the input. The experiment retains full-rank affine candidates rather than imposing a rank restriction to avoid this behavior.

3.3 Raw reconstruction spans and the Koopman connection

For this comparison only, fix a single local geometry and one *common* scalar gate $g(z)$. Suppress the local index, put $n = n_j$, and write $Q, G, C = (G + \lambda I_n)^{-1}, P, \mu$ for its references, Gram matrix, inverse, basis, and mean, with $\lambda > 0$. Let $\mathcal{S}_0$ be the real linear span of g and the coordinates of $g(z)P^\top(z - \mu)$. Define $d_m = g(\mathcal{A}^m - \mathcal{A}^A)$ for $m = K, R$. Adjoining a vector to a span means adjoining all its scalar coordinate functions.

Proposition 3 (Equal augmented raw spans)**.** *Suppose $G\mathbf{1}_n = \kappa\mathbf{1}_n$ and $C\mathcal{V} \subseteq \mathcal{V}$, where $\mathcal{V} = \operatorname{span}\{\mathbf{1}_n, Q_{:1}, \ldots, Q_{:d}\} \subseteq \mathbb{R}^n$ and $Q_{:i}$ is column i of Q. Then*

$$\operatorname{span}(\mathcal{S}_0, d_K) = \operatorname{span}(\mathcal{S}_0, d_R).$$

Adjoining a common intercept preserves the equality.

The assumption concerns a subspace of reference-coordinate vectors, not a state-transition law. It includes symmetric two-reference examples for which the reconstruction formulas differ but their augmented spans agree. It does not hold for every bank.

Proposition 4 (An irregular scalar geometry)**.** *Let $q_1 < q_2 < q_3$ with $q_1 + q_3 \neq 2q_2$, define $k_i(z) = \exp(-\beta(z - q_i)^2/2)$ for $\beta > 0$, and let $G_{ij} = \exp(-\beta(q_i - q_j)^2/2)$. Set $C = (G + \lambda I_3)^{-1}$ for $\lambda > 0$ and suppose $C\mathbf{1}_3$ is entrywise positive. The reconstruction maps use $Q = (q_1, q_2, q_3)^\top$ in Equation (2). For $g = 1$ and $\mathcal{A}^A(z) = z$, as scalar functions on $\mathbb{R}$,*

$$\mathcal{A}^K \notin \operatorname{span}(1, z, \mathcal{A}^R), \qquad d_K \notin \operatorname{span}(1, z, d_R).$$

The appendix gives an explicit admissible example. These propositions delimit raw expressivity; neither orders regularized risks. In particular, they do not compare Γ^K and Γ^R, whose gates generally differ and whose scores are nonlinear functions of reconstruction.

For a specified deterministic evolution $F : \mathcal{Z} \to \mathcal{Z}$, the Koopman operator acts on scalar functions as $\mathcal{U}_F f = f \circ F$. Exact linear coordinate evolution requires invariance of their coordinate-function span, as discussed by Kumar & Kargaran (2026). Equal raw spaces have the same invariance status under this fixed operator. Proposition 4 permits different candidate spaces but establishes invariance of neither. Moreover, an injective encoding admits a conjugated evolution on its image, but that evolution need not be linear. The observed-history forecasting experiment below does not assume a deterministic Markov state or fit a transition matrix. These qualifications prevent a direct-output gain from being read as evidence of autonomous closure.

4 Matched predictive evaluation

4.1 Finite decoders

For each encoding Γ, fit its column means and population standard deviations on A. Scales below 10^{-8} are replaced by one; no column is deleted. Let $u(z) \in \mathbb{R}^p$ denote the resulting standardized encoding.

Membership-prefix transformations are identical when an enrichment is appended at the same reconstruction and cap. There is no radial bounding map and no forecast clipping.

The linear decoder uses $f(z) = u(z)$. The nonlinear decoder retains this input and appends a Gaussian Nyström block. Choose 64 A landmark row identities, shared across all encodings and caps. Their encoded values $\ell_i \in \mathbb{R}^p$ differ between representations. For bandwidth $\sigma > 0$, let $k_\sigma(u, v) = \exp(-\|u - v\|_2^2/(2\sigma^2))$. If the landmark Gram matrix has retained eigenvectors V and positive eigenvalues in the diagonal matrix Λ, let $k_L(u)$ be the column vector of its 64 landmark similarities. The nonlinear block is $\nu(u) = \Lambda^{-1/2}V^\top k_L(u)$. After its own A-only column standardization, denoted by $\widetilde{\nu}$, the decoder input is $f(z) = \mathrm{col}(u(z), \widetilde{\nu}(u(z)))$. The supplement fixes the bandwidth scale and eigenvalue thresholds. Equal landmark counts do not equate effective rank, feature width, or computation.

Let n_A be the number of retained A rows, let $H = |\mathcal{H}|$, let $Y_i \in \mathbb{R}^H$ contain the A targets, and let p_f be the width of f. Fit $a \in \mathbb{R}^H$ and $W \in \mathbb{R}^{p_f \times H}$ by

$$(\widehat{a}, \widehat{W}) = \arg\min_{a,W} \left\{ \frac{1}{n_A} \sum_{i \in A} \|Y_i - a - W^\top f(z_i)\|_2^2 + \alpha \|W\|_F^2 \right\}, \quad \alpha > 0. \tag{8}$$

The intercept is unpenalized. One penalty, and one bandwidth for the nonlinear decoder, is selected jointly across horizons; the coefficient columns remain horizon-specific. Forecasts are direct evaluations of $\widehat{a} + \widehat{W}^\top f(z_t)$, not powers of a common transition matrix.

4.2 Laws, allocation, and candidate selection

The study has 64 independent repetitions under each of four numerical observation laws: linear autoregression, nonlinear autoregression, Van der Pol, and Duffing. The oscillators include sensor noise. Supplement Section S1 provides every equation, initialization, burn-in, and integration convention. The laws were used in development studies; fresh repetitions are not new application domains.

Each task contains independently generated A/B/C/D sets with 64/32/32/128 episodes of length 72. Histories have $d = 12$ and $\mathcal{H} = \{1, 4, 12\}$. Zero-based origins $11, 15, \ldots, 59$ give thirteen windows per episode. A/B/C retain 512/256/256 uniformly sampled windows without replacement; D retains all 1,664. All procedures share the task's rows. Windows within an episode overlap and are not separate inferential units.

Two A-fitted k-means partitions request 2 and 4 groups, with at most 24 references per group. Native cap candidates are $1, 2, 4, 8, 12$. The ridge grid is $\{10^{-6}, 10^{-5}, \ldots, 10^{2}\}$ and the nonlinear bandwidth multipliers are $0.25, 0.5, 1, 2, 4$. The affinity parameters are fixed at $\tau = 10^{-6}$ and $\omega = 16$. Every candidate is fitted on A. B minimizes the mean horizon MSE divided by the corresponding retained-A target population variances, each floored at 10^{-12}. Exact ties use ascending cap, bandwidth, and penalty order. There is no A+B refit.

In the *primary* view, every complete reconstruction–encoding–decoder procedure selects its own cap and settings. The raw-input control has no cap search. In the *secondary* view, affine-local linear ridge selects one common cap; each procedure then selects its own within-cap decoder settings. This is an asymmetric selector, not a neutral bank choice. The primary question concerns deployable selected procedures; the secondary question conditions on a specified common bank. Neither isolates the causal effect of adding a block with every fitted choice fixed.

The sample size, menu, and analysis were fixed locally before production data generation, with knowledge of development results. All A/B fits and selections were sealed before C/D generation, and all C/D predictions before scoring. C supplies diagnostics only. Source hashes document execution order, not external oversight or cryptographic target isolation. No candidate, task, or comparison was excluded. The study is not described as publicly preregistered.

4.3 Loss, estimand, and the complete primary family

For law ℓ, repetition r, procedure m, and horizon h, let $v_{\ell rh}$ be the retained-A target variance with population divisor and floor 10^{-12}. Let y_{ih} and $\widehat{y}_{mih}$ be the D target and forecast. Define standardized mean squared

error (SMSE) by

$$L_{\ell rm} = \frac{1}{3} \sum_{h \in \mathcal{H}} \frac{1}{1664} \sum_{i \in D} \frac{(y_{ih} - \widehat{y}_{mih})^2}{v_{\ell rh}}. \tag{9}$$

The estimand is expected terminal loss over the stated finite training, validation, reference-selection, and evaluation procedure for one law and decoder. The random A denominator is part of that procedure. This is not population-optimal decoder risk or a ratio of population expected errors.

Five comparisons are made for every law and decoder: KAHM-local against KAHM-memberships, KAHM-normalizer against KAHM-memberships, and KAHM-local against affine-local, positive-kernel-local, and raw input. These $5 \times 4 \times 2 = 40$ comparisons form one two-sided confidence family. For a candidate a and reference b, put $d_r = L_{\ell ra} - L_{\ell rb}$, $\bar{d} = 64^{-1} \sum_r d_r$, and $s_d^2 = 63^{-1} \sum_r (d_r - \bar{d})^2$. The reported interval is

$$\bar{d} \ \pm \ t_{63,\,1-0.05/(2\cdot 40)} \frac{s_d}{8}, \qquad t_{63,0.999375} = 3.37962646388, \tag{10}$$

where $t_{\nu,q}$ is the Student quantile with ν degrees of freedom and probability q. A wholly negative interval identifies lower candidate mean loss; a wholly positive interval identifies higher loss. Otherwise the direction is unresolved. A zero sample variance receives no Student interval rather than automatic success. No such exception changes the family size.

Bonferroni adjustment targets simultaneous 95% coverage when the marginal intervals are valid. Their validity relies on a paired-Student approximation for independent repetition differences and is not distribution-free. No practical, noninferiority, or equivalence margin is specified. Relative gain is $100(1 - \overline{L}_a/\overline{L}_b)$, a ratio of mean losses, not a confidence endpoint or an average of repetitionwise percentages. Laws are not pooled into a primary aggregate. The 64-repetition budget was the first of $32, 48, 64, 96$ giving a simultaneous half-width below $0.45 s_d$; it provides no guarantee of useful absolute precision or resolution of small effects.

5 Results

5.1 Enrichment and matched controls answer different questions

Table 1 displays every primary decision, and Table 2 gives the local-feature and raw-input losses. The supplement reports all 40 adjusted intervals and all representation means, including memberships, normalizers, and full scores. The counts of resolved comparisons are a description of this family, not a prespecified omnibus success statistic.

Local-position enrichment reduces KAHM membership-only mean loss in six of eight law–decoder cells. The two nonlinear-autoregression comparisons remain unresolved. Normalizer enrichment reduces mean loss in three cells: both Van der Pol decoders and Duffing with linear ridge. Its other comparisons remain unresolved. These results concern independently selected complete procedures, so their gains include changes in selected caps and decoder settings.

The reconstruction controls substantially narrow the interpretation. KAHM-local has lower loss than affine-local in one cell, higher loss in four, and unresolved ordering in three. Against positive-kernel-local, it has lower loss in three linear-decoder cells and unresolved ordering in the other five. In particular, none of the four nonlinear-decoder KAHM/positive-kernel comparisons resolves the ordering. Against raw inputs, three comparisons favor KAHM, three favor raw inputs, and two remain unresolved.

5.2 A favorable setting and its limits

On Van der Pol with the nonlinear decoder, KAHM-local has mean SMSE 0.002145667, compared with 0.005660276 for raw histories and 0.002200870 for affine-local features. The gains of 62.09% and 2.51% have adjusted intervals below zero in absolute-loss units. Positive-kernel-local has mean SMSE 0.002146480. Its approximately 0.038% difference from KAHM is unresolved. The large raw-input improvement is therefore not evidence of a similarly large KAHM-specific benefit.

Table 1: Complete primary decision family. "Lower" and "Higher" indicate that the adjusted interval for candidate minus reference mean SMSE is wholly negative or positive. "Unresolved" is not equivalence. The candidate is KAHM-local except in Norm/Mem, where it is KAHM-normalizer. Mem, Aff, Pos, and Raw denote KAHM-memberships, affine-local, positive-kernel-local, and raw input. All entries use independent validation selection and 64 repetitions within the named law.

Law	Decoder	Local/Mem	Norm/Mem	Local/Aff	Local/Pos	Local/Raw
Linear AR	Linear	Lower	Unresolved	Higher	Lower	Higher
Linear AR	Nonlinear	Lower	Unresolved	Higher	Unresolved	Higher
Nonlinear AR	Linear	Unresolved	Unresolved	Unresolved	Unresolved	Unresolved
Nonlinear AR	Nonlinear	Unresolved	Unresolved	Unresolved	Unresolved	Unresolved
Van der Pol	Linear	Lower	Lower	Higher	Lower	Lower
Van der Pol	Nonlinear	Lower	Lower	Lower	Unresolved	Lower
Duffing	Linear	Lower	Lower	Unresolved	Lower	Lower
Duffing	Nonlinear	Lower	Unresolved	Higher	Unresolved	Higher

Table 2: Terminal mean SMSE for local encodings and raw inputs under independent selection. Lower is better. The nonlinear decoder has 64 Gaussian Nyström landmarks and retains its linear input. Mean ordering alone does not give the adjusted decisions in Table 1.

Law	Decoder	Raw	Affine-local	Positive-local	KAHM-local
Linear AR	Linear	0.9128229	0.913372	0.9335016	0.9308015
Linear AR	Nonlinear	0.9302499	0.9175875	0.9356537	0.9342267
Nonlinear AR	Linear	0.02195891	0.01992754	0.0205418	0.02080781
Nonlinear AR	Nonlinear	0.02016172	0.02048062	0.02069077	0.02186422
Van der Pol	Linear	0.07521666	0.04048115	0.05706918	0.0497702
Van der Pol	Nonlinear	0.005660276	0.00220087	0.00214648	0.002145667
Duffing	Linear	0.01933101	0.01861734	0.01878376	0.01842372
Duffing	Nonlinear	0.0110005	0.01125626	0.01319765	0.01352628

The linear-decoder comparison illustrates why no single reconstruction ranking follows. On Van der Pol, KAHM-local has 12.79% lower mean loss than positive-kernel-local, but 22.95% higher loss than affine-local. Both adjusted comparisons resolve their respective directions. Beating one reconstruction control does not establish that KAHM is the best construction. Neither this contrast nor the nonlinear result is a separately tested representation-by-capacity interaction.

5.3 Adverse outcomes and uncertainty

On Duffing with the nonlinear decoder, KAHM-local has 22.96% higher mean loss than raw input and 20.17% higher loss than affine-local; both comparisons resolve in the unfavorable direction. On linear autoregression, affine-local and raw input outperform KAHM-local with both decoders. All ten primary nonlinear-autoregression comparisons remain unresolved.

The latter result is not a statement that the procedures behave identically. For nonlinear autoregression with the nonlinear decoder, normalizer enrichment is worse than memberships in 63 of 64 repetitions, yet the adjusted interval for the mean difference contains zero. The difference sequence has a large adverse observation and substantial dispersion. Conversely, local enrichment improves on memberships in 61 repetitions but its adjusted mean-loss interval is also unresolved. Sign counts concern frequency of improvement; the primary intervals concern the mean difference. No observation was trimmed, and no sign test or alternative interval replaced the specified analysis. The skewed differences make its approximation-based coverage a substantive limitation. The complete differences and sensitivity summaries are supplied.

5.4 Selection and information-equivalent encodings

Five of the 40 point orderings reverse between independent selection and the secondary common-cap view. The latter is conditional on a bank chosen for affine-local linear ridge and carries no additional inferential family. A favorable secondary ordering cannot rescue an unsuccessful primary comparison. For example, the nonlinear Duffing KAHM/positive-kernel point ordering changes sign, while its independently selected comparison remains unresolved.

At the same reconstruction and cap, the full-score and normalizer-enriched encodings determine one another by Proposition 1. Their fitted predictions nevertheless differ. The supplement reports all 24 same-bank mean differences across the three reconstructions, four laws, and two decoders. This comparison illustrates decoder access under different parameterizations; it does not establish that one member of an information-equivalent pair contains more information. Independently selected caps can change the score map itself and do not support the same interpretation.

5.5 Numerical behavior and computational scope

The observed stacked local bases have full numerical column rank in 254 of 256 cap-two banks and all cap-four, cap-eight, and cap-twelve banks. No cap-one stack has full column rank. These checks concern the specified numerical tolerance and sampled banks, not uniform inverse conditioning. Across all evaluated queries, the smallest recorded memberships are of order 10^{-44} for KAHM and 10^{-46} for positive-kernel reconstruction. Thus exact positivity should not be confused with a useful numerical lower bound.

The retained computational audit checks all 256 tasks, independently refits 843,264 candidate ridge heads, replays 62,464 C/D forecast arrays, and recomputes all primary statistics. The maximum absolute discrepancy in its independent B forecasts is 3.84×10^{-8} under componentwise tolerances 2×10^{-7} relative and 2×10^{-9} absolute. The audit shares the feature encoder and native routines and does not independently reimplement the generators or clustering. It is computational verification, not external replication. Supplement Sections S2 and S4 specify its scope, numerical recoveries, underflow diagnostics, and complete-search resource measurements.

6 Interpretation and limitations

The motivating problem is that geometric affinity alone does not establish predictive usefulness. The study separates information retained by an enrichment from the forecasting performance of a specified decoder and from a general preference for one reconstruction formula. Local coordinates can preserve position that a reconstruction score obscures. Their inverse relation explains available information, but the forecasts depend additionally on finite samples, feature scaling, selected dimensions, and regularization. Matched affine and positive-kernel controls are therefore essential: the large improvement over memberships alone is not a KAHM-specific comparison.

Earlier controlled studies examined related questions under different rank policies and decoders. A fixed-input branch intervention distinguishes reconstruction from target-relevant position. A separate 24-repetition capacity study reports favorable Van der Pol outcomes and unfavorable nonlinear-autoregression and Duffing comparisons. Supplement Section S5 gives their complete comparison families and numerical conventions. These observations are not pooled with the present cohort.

Different downstream architectures also limit application claims. A dimension-selected residual forecaster did not establish superiority over a selected non-KAHM policy, and a fresh policy-augmentation study failed both specified nonlinear superiority margins. On a descriptive panel of 48 M4 training series, KAHM augmentation increased aggregate loss by 2.273%. These procedures differ from the current score-based readouts. Their adverse findings remain in the supplement, but are not presented as tests of the same representation.

The main evaluation has four known laws, one history width, three horizons, and a single nonlinear landmark budget. There is no real-series evaluation of this exact procedure and no comparison against a comprehensive roster of modern forecasting systems. The goal is a controlled representation comparison, not state-of-the-art forecasting. Nominal paired-Student uncertainty, bounded candidate grids, numerical sensitivity of the inverse

encoding, and the absence of an autonomous transition model further limit the conclusions. A larger future grid or a different target population would define a new procedure, not retroactively change these results.

7 Conclusion

KAHM reconstruction-score soft coordinates admit two distinct enrichments: retaining partition normalizers recovers absolute score information, while gated local position can preserve the input under explicit rank and positivity conditions. Neither fact alone determines fitted prediction accuracy. The matched evaluation finds useful local enrichment in several settings but conditional and sometimes unfavorable comparisons with affine, positive-kernel, and raw-input controls. Raw reconstruction-span analysis provides a limited connection to candidate Koopman observable spaces; it does not establish closure of the learned score coordinates. The resulting account identifies what the representation retains, how a specified decoder can use it, and where the evidence does not support a broader KAHM advantage.

Reproducibility and tool use. The supplementary material specifies the complete numerical procedure and the full primary comparison family. The accompanying source includes repetition-level losses, analysis scripts, source identities, and an inventory of the separately stored model and episode arrays. Compiling the documents, recomputing saved-loss statistics, replaying models, and regenerating the cohort are distinguished. AI tools assisted protocol development, implementation, numerical checking, analysis, and manuscript preparation. The authors remain responsible for the scientific content and final verification.

References

Stefano De Marchi, Andrea Idda, and Gabriele Santin. A rescaled method for RBF approximation. In Gregory E. Fasshauer and Larry L. Schumaker (eds.), *Approximation Theory XV: San Antonio 2016*, pp. 39–59. Springer, 2017. doi: 10.1007/978-3-319-59912-0_3. URL https://doi.org/10.1007/978-3-319-59912-0_3.

J. Doyne Farmer and John J. Sidorowich. Predicting chaotic time series. *Physical Review Letters*, 59(8): 845–848, 1987. doi: 10.1103/PhysRevLett.59.845.

Mohit Kumar and Somayeh Kargaran. Prediction limits and Koopman closure of geometry-induced soft state abstractions. arXiv:2609.32652v3, 2026. URL https://arxiv.org/abs/2609.32652v3.

Mohit Kumar, Bernhard A. Moser, and Lukas Fischer. On mitigating the utility-loss in differentially private learning: A new perspective by a geometrically inspired kernel approach. *Journal of Artificial Intelligence Research*, 79:515–567, 2024. doi: 10.1613/jair.1.15071. URL https://jair.org/index.php/jair/article/view/15071.

Mohit Kumar, Alexander Valentinitsch, Magdalena Fuchs, Mathias Brucker, Juliana Bowles, Adnan Husakovic, Ali Abbas, and Bernhard A. Moser. Geometrically inspired kernel machines for collaborative learning beyond gradient descent. *Journal of Artificial Intelligence Research*, 83, 2025. doi: 10.1613/jair.1.16821. URL https://jair.org/index.php/jair/article/view/16821. Article 16, 35 pages.

Bethany Lusch, J. Nathan Kutz, and Steven L. Brunton. Deep learning for universal linear embeddings of nonlinear dynamics. *Nature Communications*, 9:4950, 2018. doi: 10.1038/s41467-018-07210-0.

Andreas Mardt, Luca Pasquali, Hao Wu, and Frank Noé. VAMPnets for deep learning of molecular kinetics. *Nature Communications*, 9:5, 2018. doi: 10.1038/s41467-017-02388-1. URL https://doi.org/10.1038/s41467-017-02388-1.

E. A. Nadaraya. On estimating regression. *Theory of Probability and Its Applications*, 9(1):141–142, 1964. doi: 10.1137/1109020. URL https://epubs.siam.org/doi/10.1137/1109020.

Christopher K. I. Williams and Matthias Seeger. Using the Nyström method to speed up kernel machines. In Todd K. Leen, Thomas G. Dietterich, and Volker Tresp (eds.), *Advances in Neural Information Processing Systems*, volume 13, pp. 682–688. MIT Press, 2001. URL https://proceedings.neurips.cc/paper/2000/hash/19de10adbaa1b2ee13f77f679fa1483a-Abstract.html. Proceedings of NIPS 2000.

Yilun Xu, Shengjia Zhao, Jiaming Song, Russell Stewart, and Stefano Ermon. A theory of usable information under computational constraints. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=r1eBeyHFDH.

Kai Yu and Tong Zhang. Improved local coordinate coding using local tangents. In *Proceedings of the 27th International Conference on Machine Learning*, pp. 1215–1222, 2010. URL https://icml.cc/2010/papers/454.pdf.

Kai Yu, Tong Zhang, and Yihong Gong. Nonlinear learning using local coordinate coding. In *Advances in Neural Information Processing Systems*, volume 22, pp. 2223–2231, 2009. URL https://proceedings.neurips.cc/paper/2009/hash/2afe4567e1bf64d32a5527244d104cea-Abstract.html.

A Proofs and explicit examples

The first two propositions concern exact representations at fixed geometry. The last two concern exact raw reconstruction functions on their stated domains. None of the proofs assumes that the evaluated forecasting representations satisfy dynamical closure.

A.1 Score recovery and positivity

Proof of Proposition 1. Write $a_j = (1 - T_j^m + \tau)^\omega$. Since $T_j^m \in [0,1]$, $\tau^\omega \le a_j \le (1+\tau)^\omega$. All affinities are positive, so their sum Z_b^m and memberships are positive. Taking logarithms in $\phi_j^m = a_j / Z_b^m$ gives $\log \phi_j^m + \zeta_b^m = \omega \log(1 - T_j^m + \tau)$. Exponentiating and solving for T_j^m proves Equation (6). Equation (4) supplies the converse map. Finally,

$$\phi_j^m = \frac{a_j}{a_j + \sum_{i \neq j} a_i} \ge \frac{\tau^\omega}{\tau^\omega + (q-1)(1+\tau)^\omega},$$

because the fraction increases with its numerator affinity and decreases with every competing affinity. The argument applies independently to every partition. □

A.2 Input recovery and a conditioning bound

Proof of Proposition 2. All memberships are supplied as part of the local encoding. Dividing its jth coordinate block by the positive membership gives $P_j^\top (z - \mu_j)$. Adding the known $P_j^\top \mu_j$ and stacking yields $q(z) = \mathcal{P} z$. Full column rank makes $\mathcal{P}^\top \mathcal{P}$ invertible and gives the displayed recovery formula, hence injectivity.

For the conditioning statement put $w_j = P_j^\top (z - \mu_j)$ and $w_j' = P_j^\top (z' - \mu_j)$. Use $v_j = \phi_j w_j$ and $v_j' = \phi_j' w_j'$, with rule and query arguments suppressed. Then

$$w_j - w_j' = \frac{v_j - v_j'}{\phi_j} + \frac{\phi_j' - \phi_j}{\phi_j} w_j'.$$

Stacking these identities and applying the triangle inequality gives $\|\mathcal{P}(z - z')\|_2 \le (\|v - v'\|_2 + M\|\phi - \phi'\|_2)/\varepsilon$. The smallest-singular-value inequality $\|\mathcal{P}(z - z')\|_2 \ge \sigma_{\min}(\mathcal{P}) \|z - z'\|_2$ proves Equation (7). The constants are assumptions on the query pair or domain; they are not automatically favorable because an inverse exists. □

A.3 Equality of augmented raw reconstruction spans

Proof of Proposition 3. A Gaussian Gram matrix is symmetric positive semidefinite. Therefore $\kappa \ge 0$ and C is symmetric positive definite. The row-sum assumption gives $C\mathbf{1}_n = a\mathbf{1}_n$ with $a = (\kappa + \lambda)^{-1} > 0$. The KAHM denominator is $a\mathbf{1}_n^\top k(z) > 0$. Invariance of $\mathcal{V}$ gives a matrix $B \in \mathbb{R}^{d \times d}$ and a vector $c \in \mathbb{R}^d$ such that $CQ = QB + \mathbf{1}_n c^\top$. Symmetry then yields

$$\mathcal{A}^K(z) = \frac{(CQ)^\top k(z)}{a\mathbf{1}_n^\top k(z)} = \frac{B^\top \mathcal{A}^R(z) + c}{a}.$$

Every coordinate of $g\mathcal{A}^A = g\mu + P\{gP^\top(z-\mu)\}$ belongs to $\mathcal{S}_0$. Multiplying the preceding identity by g therefore gives $\operatorname{span}(\mathcal{S}_0, d_K) \subseteq \operatorname{span}(\mathcal{S}_0, d_R)$.

The restriction of C to $\mathcal{V}$ is injective. Because $\mathcal{V}$ is finite dimensional and $C\mathcal{V} \subseteq \mathcal{V}$, it follows that $C\mathcal{V} = \mathcal{V}$. Thus there are $D \in \mathbb{R}^{d\times d}$ and $e \in \mathbb{R}^d$ such that $C^{-1}Q = QD + \mathbf{1}_n e^\top$. Multiplying by C gives $Q = CQD + a\mathbf{1}_n e^\top$, so

$$\mathcal{A}^R(z) = aD^\top \mathcal{A}^K(z) + ae.$$

The same gated-span argument gives the reverse inclusion. No independence of the columns of Q, or invertibility of B or D, is required. Adjoining the same constant function preserves equality. □

For scalar references $(-1, 1)$, let $\rho \in (0, 1)$ be the off-diagonal Gaussian Gram entry. With $g = 1$, $P = 1$, and $\mathcal{A}^A(z) = z$, direct inversion gives

$$\mathcal{A}^K = c_\rho \mathcal{A}^R, \qquad d_K = c_\rho d_R + (c_\rho - 1)z, \qquad c_\rho = \frac{1+\rho+\lambda}{1-\rho+\lambda}.$$

The departure formulas differ, but the augmented spaces containing z coincide. This example concerns jointly variable linear coefficients; freezing a separate common forecast would define a different comparison.

A.4 Separation on an irregular scalar reference set

Proof of Proposition 4. Put $q = (q_1, q_2, q_3)^\top$, $v = C\mathbf{1}_3$, and $u = Cq$. Symmetry of C gives $\mathcal{A}^K = u^\top k/(v^\top k)$ and $\mathcal{A}^R = q^\top k/(\mathbf{1}_3^\top k)$. Since $v_i > 0$, the first function is a convex combination of the fixed values u_i/v_i, with weights proportional to $v_i k_i(z)$. The second is a convex combination of the references. Both are bounded on $\mathbb{R}$.

Suppose first that $\mathcal{A}^K = a\mathcal{A}^R + b$ for constants a, b. Let $w = u - bv$. Cross multiplication, with positive denominators, gives

$$(w^\top k)(\mathbf{1}_3^\top k) = a(q^\top k)(v^\top k). \tag{11}$$

For $i \le j$,

$$k_i(z)k_j(z) = e^{-\beta z^2} e^{\beta(q_i+q_j)z} e^{-\beta(q_i^2+q_j^2)/2}.$$

The six sums $q_i + q_j$ are distinct: strict ordering rules out every possible equality except $q_1 + q_3 = 2q_2$, which is excluded. Exponentials with distinct real rates are linearly independent. To see this, differentiate a proposed zero linear combination at zero through orders zero to five. Its coefficients satisfy a Vandermonde system with nonzero determinant. Multiplication by the common positive factor $e^{-\beta z^2}$ and the nonzero constants preserves independence.

Comparing coefficients in Equation (11) therefore gives $w_i = aq_i v_i$ and $w_i + w_j = a(q_i v_j + q_j v_i)$ for $i < j$. If $a = 0$, then $w = 0$ and invertibility of C implies $q = b\mathbf{1}_3$, contradicting distinct references. Otherwise substitution gives $(q_i - q_j)(v_i - v_j) = 0$, so all entries of v equal a positive value. The identity $(G + \lambda I_3)v = \mathbf{1}_3$ would then force constant row sums in G.

But its middle row exceeds its first row by $G_{23} - G_{13} > 0$ and its last row by $G_{12} - G_{13} > 0$: each neighboring reference distance is shorter than $q_3 - q_1$. Thus constant row sums are impossible. This rules out an affine relation between the two reconstructions. A relation $\mathcal{A}^K = a\mathcal{A}^R + b + cz$ forces $c = 0$ by boundedness, so it is also impossible. Finally, $d_m = \mathcal{A}^m - z$ gives $\operatorname{span}(1, z, d_R) = \operatorname{span}(1, z, \mathcal{A}^R)$, and membership of d_K in this space would imply membership of $\mathcal{A}^K$. □

An explicit positive-denominator example. Take $q = (-1, 0, 2)^\top$, $\beta = 3/7$, and $\lambda = 4$. Write $G = I_3 + E$ and $a = 5$. The maximum absolute row-sum norm satisfies $\|E\|_\infty = e^{-3/14} + e^{-6/7} < 2 < a/2$. The convergent Neumann series gives

$$\|aC\mathbf{1}_3 - \mathbf{1}_3\|_\infty \le \frac{\|E\|_\infty/a}{1 - \|E\|_\infty/a} < 1.$$

Here the vector infinity norm is the largest absolute coordinate. Consequently all entries of $C\mathbf{1}_3$ are positive. This is a constructed algebraic example; its chosen regularizer does not assert that the experiment's reference-fitted regularizers satisfy the same condition.